

ETHICAL HACKING INTERNSHIP

Task 02

Network Scanning & Service Enumeration

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Target: localhost 127.0.0.1

Date: June 27, 2026

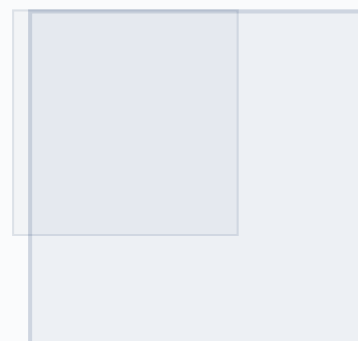


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Part A: Nmap Installation

Tool: Nmap 7.99

Platform: i686-pc-windows-windows

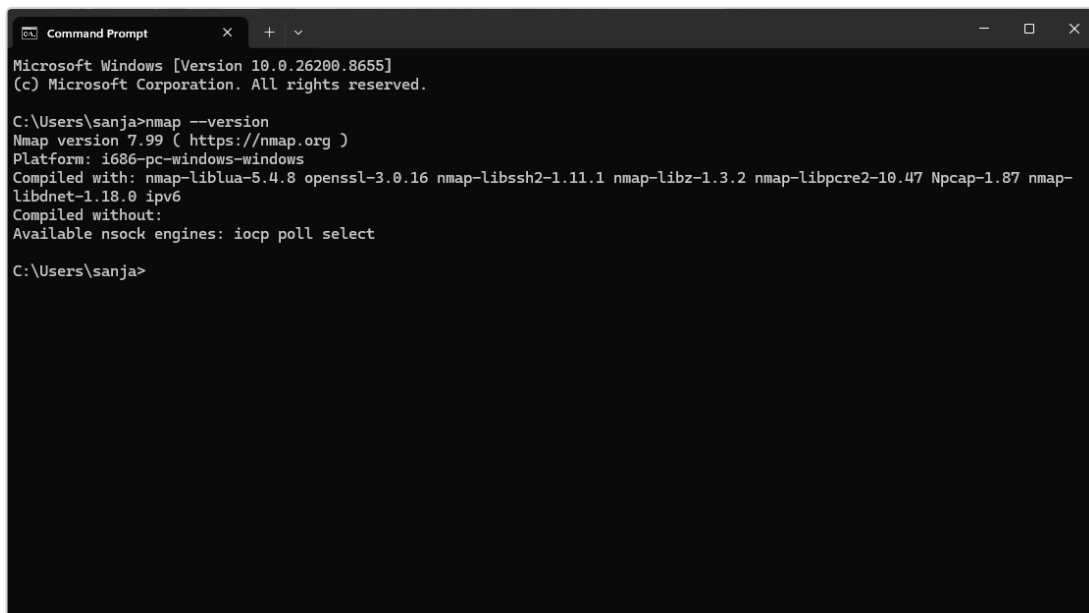
Npcap Version: 1.87

Installation Source: <https://nmap.org/download.html>

Verification command used:

```
nmap --version
```

Screenshot



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sanja>nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcap-1.0.47 Npcap-1.87 nmap-
libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Users\sanja>
```

Figure 1: `nmap --version` — confirms Nmap 7.99 installed successfully on Windows 11

Part B: Basic Localhost Scan

Command used:

```
nmap localhost
```

Field	Value
Target	localhost (127.0.0.1)
Host Status	Up (0.00019s latency)
Total Ports Scanned	1000
Closed Ports	998
Open Ports	2
Scan Time	0.33 seconds

Open Ports Found

Port	State	Service
135/tcp	open	msrpc
445/tcp	open	microsoft-ds

HIGH RISK

Screenshot

```

Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sanja>nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcap-1.0.47 Npcap-1.87 nmap-
libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Users\sanja>nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 13:34 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00019s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrcpc
445/tcp    open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.33 seconds
C:\Users\sanja>

```

Figure 2: `nmap localhost` — 2 open ports found: 135 (`msrpc`) and 445 (`microsoft-ds`)

Part C: Service Version Detection

Command used:

```
nmap -sV localhost
```

Port	Service	Version
135/tcp	msrpc	Microsoft Windows RPC
445/tcp	microsoft-ds	Not fully detected (microsoft-ds?)

Additional info detected:

- Service Info: OS: Windows
- CPE: `cpe:/o:microsoft:windows`
- Service detection time: 7.00 seconds

Screenshot

```
Command Prompt
Compiled without:
Available nsock engines: iocp poll select

C:\Users\sanja>nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 13:34 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00019s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrcpc
445/tcp    open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.33 seconds

C:\Users\sanja>nmap -sV localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 13:34 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000085s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrcpc       Microsoft Windows RPC
445/tcp    open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.00 seconds

C:\Users\sanja>
```

Figure 3: `nmap -sV localhost` — service version detection reveals Microsoft Windows RPC on port 135

Part D: Operating System Detection

Command used:

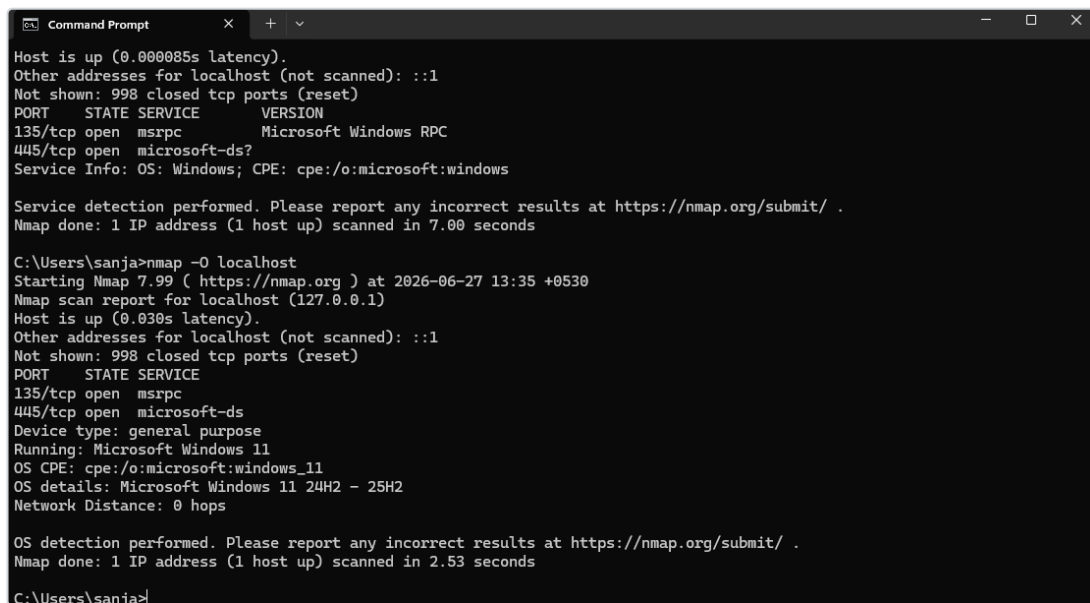
```
nmap -O localhost
```

Field	Value
OS Detected	Yes
Device Type	General Purpose
Running	Microsoft Windows 11
OS CPE	cpe:/o:microsoft:windows_11
OS Details	Microsoft Windows 11 24H2 – 25H2
Network Distance	0 hops

Why is OS Detection Useful?

- Helps identify OS-specific vulnerabilities and unpatched CVEs.
- Allows penetration testers to select the right exploits for the target OS.
- Reveals the Windows version — older builds may have known critical vulnerabilities.
- Helps understand the likely default services and configurations running on the machine.

Screenshot



```
Command Prompt
Host is up (0.000085s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
445/tcp   open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.00 seconds

C:\Users\sanja>nmap -O localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 13:35 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.030s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
445/tcp   open  microsoft-ds
Device type: general purpose
Running: Microsoft Windows 11
OS CPE: cpe:/o:microsoft:windows_11
OS details: Microsoft Windows 11 24H2 - 25H2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.53 seconds

C:\Users\sanja>
```

Figure 4: `nmap -O localhost` — OS detected as Microsoft Windows 11 24H2–25H2

Part E: Common Port Research

Port	Protocol	Purpose	Common Security Risks
20/21	FTP	File transfer between client and server	Plaintext transmission, anonymous login, brute force
22	SSH	Encrypted remote login and command execution	Brute force on weak passwords, outdated SSH versions
23	Telnet	Unencrypted remote login	Everything sent in plaintext including passwords — highly insecure
25	SMTP	Sends emails between mail servers	Open relay abuse for spam, email spoofing without SPF/DKIM
53	DNS	Resolves domain names to IP addresses	DNS poisoning, DNS amplification DDoS attacks
80	HTTP	Serves unencrypted web pages	MITM attacks, credential sniffing, no data encryption
110	POP3	Downloads emails from server to client	Plaintext credentials unless SSL/TLS used
143	IMAP	Access emails stored on mail server	Plaintext unless IMAPS (993) — vulnerable to sniffing
443	HTTPS	Encrypted web traffic over TLS/SSL	Misconfigured SSL, outdated ciphers, certificate spoofing
445 HIGH	SMB	Windows file sharing and network services	EternalBlue/WannaCry ransomware exploit — critical risk
3389 MED	RDP	Remote graphical desktop access to Windows	Brute force, BlueKeep CVE-2019-0708 — never expose to internet

Part F: Scan Analysis

1. Which services are currently running?

Two services were found running on `localhost`:

- Port `135` — Microsoft RPC (`msrpc`): Used by Windows for inter-process communication between local and remote processes.
- Port `445` — SMB (`microsoft-ds`) **HIGH RISK**: Used for Windows file and printer sharing across the network.

2. Are all open ports necessary?

Port `135` (RPC) is a core Windows system requirement and cannot easily be disabled without breaking system functionality. Port `445` (SMB) is necessary only if file sharing is actively used. If no file sharing is needed on this machine, port `445` should be restricted via firewall rules.

3. Which services could become security risks if misconfigured?

- Port `445` (SMB) **HIGH RISK** — This is a critical risk. The EternalBlue exploit (MS17-010) targeted SMB and was used in the devastating WannaCry ransomware attack in 2017 that affected hundreds of thousands of machines worldwide.
- Port `135` (RPC) — Has historical critical vulnerabilities such as MS03-026 (DCOM RPC vulnerability), which allowed remote code execution.

CRITICAL WARNING — SMB Vulnerability Alert

Port `445` (**SMB**) is the most significant security finding in this scan. This port was the primary attack vector for the **WannaCry ransomware** (May 2017) which exploited the **EternalBlue** vulnerability (MS17-010). EternalBlue is a exploit developed by the NSA and later leaked by the Shadow Brokers hacker group. It targets the SMBv1 protocol and allows remote code execution without authentication. Even on a fully patched Windows 11 system, SMB should be disabled or firewalled if file sharing is not actively required. **Never expose port 445 to the public internet under any circumstances.**

4. Which port would you disable if not required?

Port `445` (SMB) **HIGH RISK** — If file sharing is not needed, SMB should be disabled or blocked at the firewall level. SMB should never be exposed to the public internet under any circumstances.

Part G: Scan Report & Conclusion

Scan Summary

Field	Details
Scan Date	June 27, 2026
Target System	<code>localhost</code> (<code>127.0.0.1</code>)
Operating System	Microsoft Windows 11 24H2 – 25H2
Nmap Version	7.99
Commands Used	<code>nmap --version</code> <code>nmap localhost</code> <code>nmap -sV localhost</code> <code>nmap -O localhost</code>
Total Open Ports	2 (<code>135/tcp</code> , <code>445/tcp</code>)
Running Services	Microsoft Windows RPC (135), SMB/microsoft-ds (445) HIGH RISK

Observations

- Only 2 out of 1000 scanned ports are open — minimal attack surface.
- Both open ports are default Windows services — expected on any Windows 11 machine.
- OS was correctly identified as Windows 11 24H2–25H2 by Nmap's fingerprinting.
- No HTTP/HTTPS servers running — machine is not acting as a web server.
- No SSH or RDP exposed — good from a remote access security perspective.

Recommendations

- Restrict port 445 (SMB) via Windows Firewall if file sharing is not actively needed.
- Ensure Windows 11 is fully updated — EternalBlue and similar SMB exploits are patched in modern Windows but updates must be current.
- Monitor port 135 (RPC) — disable unnecessary DCOM services where possible.
- Run periodic Nmap scans to detect any newly opened ports or unexpected services.

Conclusion

The network scanning phase on my local Windows 11 machine using Nmap 7.99 revealed two open ports: 135 (Microsoft Windows RPC) and 445 (SMB/microsoft-ds). Both are default Windows services and expected to be present on any Windows 11 installation. The OS was correctly fingerprinted as Microsoft Windows 11 24H2-25H2, demonstrating Nmap's OS detection accuracy.

From a security perspective, port 445 (SMB) **HIGH RISK** is the most significant finding — it has been the attack vector for some of the most damaging ransomware attacks in history. While the current Windows 11 installation is patched, SMB should be restricted via firewall rules if file sharing is not actively required. Port 135 (RPC) is a Windows core requirement and difficult to disable, but unnecessary DCOM services should be minimized.

This exercise showed how an attacker — or a defender — can quickly map a machine's running services and identify potential attack surfaces using simple Nmap commands. Scanning and enumeration are the foundation of any security assessment.

Disclaimer

All scans were performed exclusively on **localhost (127.0.0.1)** — my own personal Windows 11 machine. No external systems, public IPs, or unauthorized machines were scanned at any point. This report is submitted as part of the **White Band Associates Ethical Hacking Internship Task 02**.